

The empty set (and friends)

The symbol that started it all:

$$\emptyset \quad \varnothing \quad \{x \in \mathbb{R} : x \neq x\} = \emptyset.$$

Compare in context:

$$A \cap B = \emptyset, \quad \bigcap_{n \in \mathbb{N}} (0, 1/n) = \emptyset, \quad \mathcal{P}(\emptyset) = \{\emptyset\}.$$

Number sets and blackboard bold

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}, \quad \mathbb{F}_p, \quad \mathbb{H}, \quad \mathbb{P}^n, \quad \mathbb{1}_A.$$

Greek alphabet

Lowercase:

$$\alpha \beta \gamma \delta \epsilon \zeta \eta \theta \vartheta \iota \kappa \lambda \mu \nu \xi \omicron \pi \varpi \rho \varrho \sigma \varsigma \tau \upsilon \phi \varphi \chi \psi \omega$$

Uppercase:

$$\Gamma \Delta \Theta \Lambda \Xi \Pi \Sigma \Upsilon \Phi \Psi \Omega$$

Calligraphic and script

$$\mathcal{A} \mathcal{B} \mathcal{C} \mathcal{D} \mathcal{E} \mathcal{F} \mathcal{G} \mathcal{H} \mathcal{I} \mathcal{J} \mathcal{K} \mathcal{L} \mathcal{M} \mathcal{N} \mathcal{O} \mathcal{P} \mathcal{Q} \mathcal{R} \mathcal{S} \mathcal{T} \mathcal{U} \mathcal{V} \mathcal{W} \mathcal{X} \mathcal{Y} \mathcal{Z}$$

Logic and set theory

$$\begin{aligned} \forall x \in X, \exists y \in Y \text{ such that } P(x, y) &\implies Q(y). \\ A \cup B, A \cap B, A \setminus B, A \subset B, A \subseteq B, A \subsetneq B, A^c, \mathcal{P}(A). \\ \neg p \vee (q \wedge r) &\iff (\neg p \vee q) \wedge (\neg p \vee r). \end{aligned}$$

Relations

$$= \neq \approx \equiv \sim \simeq \asymp \propto \leq \geq \ll \gg \prec \succ \preceq \succeq \perp \parallel.$$

Operators and big operators

$$\begin{aligned} \sum_{i=1}^n i &= \frac{n(n+1)}{2}, \quad \prod_{p \leq N} \left(1 - \frac{1}{p^s}\right)^{-1} = \sum_{n=1}^{\infty} \frac{1}{n^s} = \zeta(s). \\ \int_{-\infty}^{\infty} e^{-x^2} dx &= \sqrt{\pi}, \quad \iint_D f(x, y) dA, \quad \oint_{\partial \Sigma} \vec{F} \cdot d\vec{r}. \\ \bigcup_{n \in \mathbb{N}} A_n, \quad \bigcap_{n \in \mathbb{N}} A_n, \quad \bigoplus_{i \in I} V_i, \quad \bigotimes_{i=1}^n H_i, \quad \bigcup, \quad \bigsqcup. \end{aligned}$$

Calculus

$$\frac{\mathrm{d}}{\mathrm{d}x} \int_a^x f(t) \, \mathrm{d}t = f(x), \quad \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0.$$

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}, \quad \nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}.$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

Showcase equations

Cauchy–Schwarz:

$$|\langle u, v \rangle|^2 \leq \langle u, u \rangle \cdot \langle v, v \rangle.$$

Binomial theorem:

$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k.$$

Euler’s identity:

$$e^{i\pi} + 1 = 0.$$

Stokes’ theorem:

$$\int_{\partial\Omega} \omega = \int_{\Omega} \mathrm{d}\omega.$$

Linear algebra

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \quad \det \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

$$\langle Ax, y \rangle = \langle x, A^\top y \rangle, \quad \|x\|_2 = \sqrt{\sum_i x_i^2}, \quad \mathrm{tr}(AB) = \mathrm{tr}(BA).$$

Big delimiters and nesting

$$\left(\sum_{i=1}^n a_i b_i \right)^2 \leq \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right).$$
$$\left\{ x \in \mathbb{R}^n : \|x - \mu\|_2^2 \leq \chi_{\alpha,n}^2 \right\}.$$

Fractions, roots, and continued fractions

$$\frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \cdots}}} = \frac{\sqrt{5} - 1}{2}.$$
$$\sqrt[n]{x}, \quad \sqrt{a + \sqrt{b + \sqrt{c}}}, \quad \frac{a+b}{c-d}, \quad \binom{2n}{n} \sim \frac{4^n}{\sqrt{\pi n}}.$$

Arrows

$\rightarrow \rightarrow \leftarrow \leftrightarrow \Rightarrow \Leftarrow \Leftrightarrow \mapsto \hookrightarrow \rightarrowtail \rightarrow \Rrightarrow \nearrow \searrow .$

Miscellaneous symbols

$\hbar, \ell, \Re, \Im, \wp, \aleph, \beth, \partial, \nabla, \infty, \angle, \triangle, \dagger, \ddagger, \star, \ast, \circ, \bullet.$

Inline-heavy paragraph

For $x \in \mathbb{R}^n$ and $\varepsilon > 0$, the open ball $B_\varepsilon(x) = \{y \in \mathbb{R}^n : \|y - x\| < \varepsilon\}$ is a basic open set in the Euclidean topology. A function $f : X \rightarrow Y$ between topological spaces is continuous if $f^{-1}(U)$ is open in X for every open $U \subseteq Y$, equivalently $\forall x \in X, \forall \varepsilon > 0, \exists \delta > 0$ such that $d_X(x', x) < \delta \implies d_Y(f(x'), f(x)) < \varepsilon$. The empty set $\emptyset$ is open and closed in every topology, and $f^{-1}(\emptyset) = \emptyset, f(\emptyset) = \emptyset$.